Mothukuru Dhanushavardhan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
Sri Chaitanya Junior College, Vijayawada	Class XII (Board of Intermediate Education, Andhra Pradesh)	2021	97.00%
Sri Chaitanya EM High School , Anantapur	Class X (Board of Secondary Education, Andhra Pradesh)	2019	10.00%

WORK EXPERIENCE

Full Stack Developer Intern | CodTech IT Solutions

- Engineered responsive **full-stack web applications** using **React.js, Node.js, Express.js** and **MongoDB**.
- Designed and integrated **RESTful APIs** enabling seamless data flow between React and Express services.
- Refactored **reusable UI components** and API logic to improve **page responsiveness** and **user experience**.

MASTER'S THESIS

ARGUS - Adaptive Robust Graph-based framework Using Semi-supervised Learning for Bitcoin Fraud Detection*(Ongoing)*

- Designing a semi-supervised fraud detection framework for Bitcoin transaction graphs using a **teacher-student CoSemiGNN** architecture on the **Elliptic++** dataset.
- Proposed **PLSALP (Pseudo-Label Sharpening via Anchor Link Prediction)** to improve pseudo-label quality when labelled fraud data is scarce, targeting **Illegal-class F1** gains.
- Integrating **Graph-Guided Label Propagation** in place of single-hop GCN soft-labelling, with large-scale experiments on a **DGX V100 GPU cluster**.

PROJECTS

Agentic Document Assistant - Multi-Step LLM Agent over PDFs - Github

- Built an **agentic AI system** that answers questions over uploaded PDFs by planning **multi-step tool calls** (document search, summarisation, calculation) instead of a single prompt.
- Implemented a **retrieval pipeline** that chunks and embeds documents into a **FAISS vector index** for semantic search.
- Developed a **FastAPI** backend and a **React** interface that streams the agent's **reasoning trace and tool calls** in real time. Containerised with **Docker** and deployed.

AI Interview Assistant - NLP-Based Interview Simulator - Github

- Built a full-stack platform that simulates **technical interviews** with **resume parsing** and **live video Q&A**.
- Integrated **speech-to-text** with **NLP keyword-based scoring** to evaluate the relevance of every answer.
- Generated an automated **performance report** with question-wise scores and **personalised feedback**.

SKILLS

Languages: C++, Python, JavaScript, SQL

Frontend: React.js, HTML, CSS, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, Flask, RESTful APIs, Socket.IO, Server-Sent Events, JWT, bcrypt

Databases: MongoDB, MySQL, Mongoose, FAISS (Vector Search)

AI/ML: PyTorch, PyTorch Geometric, Graph Neural Networks, Semi-Supervised Learning, LLM APIs, LLM Function Calling, RAG, Embeddings, Prompt Engineering, NLP, Speech-to-Text

Tools & DevOps: Git, GitHub, Docker, Postman, Linux, VS Code

RELEVANT COURSES

Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Deep Learning

POSITIONS OF RESPONSIBILITY

Technical Events Coordinator | PIXEL 2K24 (College Fest)

- Led the **technical track** of the department's annual fest, coordinating **paper presentation, competitive coding and debugging** events for **300+ participants** from multiple colleges.
- Managed **problem-set preparation, event scheduling, volunteer allocation** and **on-day execution**, running every technical event on schedule.

CERTIFICATIONS

- Developing Front End Apps with React- edX
- Prompt Design in Vertex AI- Google Cloud
- Cloud Computing- NPTEL

SCHOLASTIC ACHIEVEMENTS

Competitive Programming: Solved **1000+ DSA problems** across **LeetCode** , **GeeksforGeeks** and other platforms

National-Level Technical E-Quiz: Secured **2nd Rank** in a contest conducted by **SVU College**.

EXTRACURRICULAR

Sports: Badminton | **Interests:** Competitive Programming